

Name:

Teacher: Martin, Ghabbar

Year 11

Mathematics Extension 1

Assessment Task 3

2016

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- In Questions 11 - 13, show relevant mathematical reasoning and/or calculations
- A reference sheet is provided with this paper

Total marks – 55

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II

45 marks

- Attempt Questions 11-13
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover
- Allow about 75 minutes for this section

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Section I

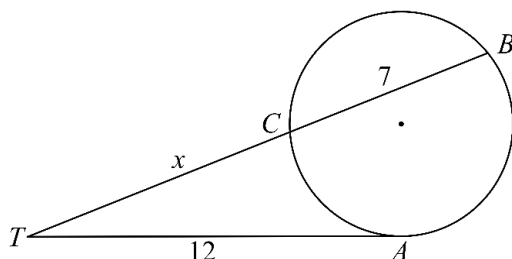
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. Line TA is the tangent to the circle at A , and TB is a secant meeting the circle at B and C .



What is the value of x given that $TA=12$ and $BC=7$?

- (A) 7
- (B) 8
- (C) 9
- (D) 10
2. How many ways can 10 boys be arranged in a line if the first boy in the line is Aidan and the last boy in the line is Zac?
- (A) 40,320
- (B) 80,640
- (C) 3,628,800
- (D) 7,257,600

3. Which of the following is equivalent to the expression $\sqrt{\frac{4+4\cos 2x}{1-\cos 2x}}$?
- (A) $2\cot^2 x$
- (B) $2\cot x$
- (C) $2\tan x$
- (D) $2\tan^2 x$
4. What is the solution to the inequality $\frac{3}{x+4} \geq 2$?
- (A) $x > -4$ and $x \leq -\frac{5}{2}$
- (B) $x \geq -4$ and $x \leq -\frac{5}{2}$
- (C) $x > -4$ and $x \geq -\frac{5}{2}$
- (D) $x \geq -4$ and $x \geq -\frac{5}{2}$
5. Find $\lim_{x \rightarrow \infty} \frac{2x^2+1}{x-1}$
- (A) x
- (B) 0
- (C) $2x$
- (D) 1
6. The point $P(-3,8)$ divides the interval AB internally in the ratio $k:1$. If A is the point $(6,-4)$ and B is the point $(0,4)$, what is the value of k ?
- (A) -1
- (B) -3
- (C) 1
- (D) 3

7. The numberplates for motor vehicles consist of 2 letters, 2 numbers and then 2 letters such as AB01CD. How many different numberplates are possible if repetition is allowed?
- (A) 135 200
- (B) 270 400
- (C) 37 015 056
- (D) 45 697 600
8. A committee of 5 is to be chosen from 9 candidates. How many different ways can this be done?
- (A) 45
- (B) 120
- (C) 126
- (D) 15 120
9. What is the value of $\lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x + 1}$?
- (A) Undefined
- (B) -1
- (C) 0
- (D) 1
10. What is the exact value of $\tan 15^\circ$?
- (A) $2 - \sqrt{3}$
- (B) $4 - \sqrt{3}$
- (C) $2 + \sqrt{3}$
- (D) $4 + \sqrt{3}$

End of section I

Section II

45 marks

Attempt Questions 11–13

Allow about 75 minutes for this section

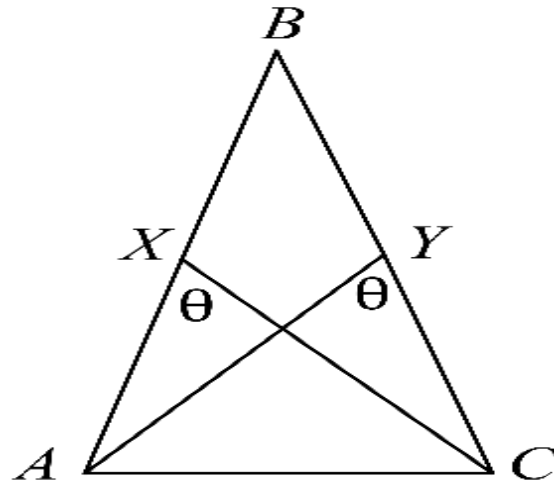
Answer each question in a **separate writing booklet**. Extra writing booklets are available.

In Questions 11–13, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- a)** How many 5 digit numbers can be formed using the digits 2, 3, 5, 6, 8, and 9:
- i.** If no digit can be used more than once? **1**
 - ii.** How many of the numbers formed are even numbers? **1**
- b)** A four-person team is chosen at random from ten women and six men.
- i.** In how many ways can this team be chosen? **1**
 - ii.** How many women only teams are possible? **1**
 - iii.** How many teams will have at least one man? **1**
- c)** Four men and four women meet for dinner at a restaurant. They are to be seated at a circular table.
- In how many different ways can they be seated if:
- i.** There are no restrictions? **1**
 - ii.** The men and women are to sit in alternate positions? **1**
 - iii.** The women all sit together, with two particular women always side by side. **2**

- d) Two angles $\angle AXC = \theta$ and $\angle AYC = \theta$ are constructed in $\triangle ABC$. **Copy the diagram onto your writing booklet.**



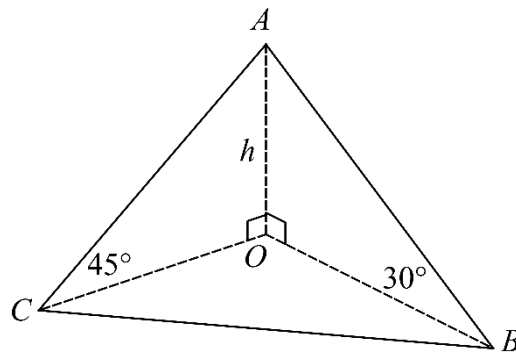
- | | | |
|-----|---|---|
| i. | Prove that AXYC is cyclic. | 1 |
| ii. | Hence show that $\angle BXY = \angle YCA$. | 1 |
-
- | | | |
|-------|--|---|
| e) i. | Express $\cos x - \sin x$ in the form $A \cos(x + \alpha)$ for $0 \leq \alpha \leq 90^\circ$. | 2 |
| ii. | Hence solve the $\cos x - \sin x = -1$ for $0^\circ \leq x \leq 360^\circ$. | 2 |

End of question 11

Question 12 (15 marks) – Start a new booklet.

a) Solve $9^x - 10 \cdot 3^x + 9 = 0$ for x . **3**

- b) The diagram below shows a vertical tower AO of height h metres, angle of elevation to the top of the tower from point B is 30° and from point C is 45° . Let $\angle BOC = 120^\circ$ and $\angle CAB = \theta$. **Copy the diagram onto your writing booklet.**



- i. Show that $OC = h$ and $OB = \sqrt{3}h$ **2**
- ii. Hence or otherwise, show that $BC^2 = (4 + \sqrt{3})h^2$. **2**
- c) i. Sketch the graph of $y = |2x - 3|$ **2**
- ii. Solve $|2x - 3| \leq |x - 3|$ **2**
- d) The point P divides the interval AB joining $A(-4, -3)$ and $B(1, 5)$ externally in the ratio 3:2. What are the coordinates of P? **2**
- e) Solve the inequality $\frac{2x+5}{x+1} < 1$ **2**

End of question 12

Question 13 (15 marks) – Start a new booklet.

- a)**
- i.** Show that the two curves $y = x^2 + 2x - 3$ and $y = x^2 + 1$ intersect at (2,5) **1**
 - ii.** What is the size of the acute angle, to the nearest degree, between the intersection of the tangents to the curves at (2,5)? **2**
- b)** Find the values of k for which the equation $4x^2 - (6+k)x + 1 = 0$ has no real zeros. **2**
- c)** Consider $f(x) = \frac{2x^2}{x^2 - 9}$
- i.** What are the equations of the vertical asymptotes? **1**
 - ii.** Describe the behaviour of $y = f(x)$ as $x \rightarrow \pm\infty$ **1**
 - iii.** Sketch $y = f(x)$, clearly showing its key features including asymptotes and x- and y-intercepts. **2**
 - iv.** State the range of this function. **1**
- d)** The line $y = mx$ is inclined at 45° to the line $2x - y - 1 = 0$. What are the values of m ? **2**
- e)** Solve $\cos 2x - 3\cos x + 2 = 0$ in the domain $0 \leq x \leq 360^\circ$ **3**

End of Exam

Mathematics Extension 1

Section I Multiple-Choice Answer Sheet

- | | | | | |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

Section I:

1. $TA^2 = TC \times TB$

$$12^2 = x \times (x+7)$$

$$144 = x^2 + 7x$$

$$x^2 + 7x - 144 = 0$$

$$(x+16)(x-9) = 0$$

$$x = -16, x = 9$$

$$\therefore x = 9 > 0 \quad (C)$$

2. $1 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320 \quad (A)$

3. $\sqrt{\frac{4+4(\cos 2x)}{1-\cos 2x}} = \sqrt{\frac{4(1+\cos 2x)}{1-\cos 2x}}$

$$= 2 \sqrt{\frac{1+(2\cos^2 x - 1)}{1-(1+2\sin^2 x)}}$$

$$= 2 \sqrt{\frac{1+2\cos^2 x - 1}{1-1-2\sin^2 x}}$$

$$= 2 \sqrt{\cot^2 x} = 2 \cot x \quad (B)$$

4. $\frac{3 \times (x+4)^2}{x+4} \geq 2 \times (x+4)^2$

$$3(x+4) \geq 2(x+4)^2$$

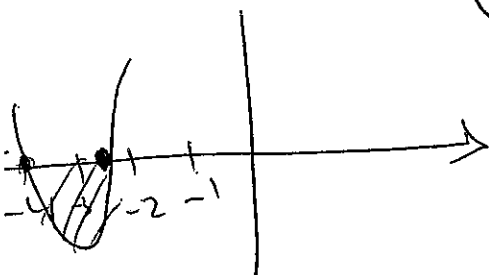
$$2(x+4)^2 - 3(x+4) \leq 0$$

$$(x+4)[2(x+4)-3] \leq 0$$

$$(x+4)(2x+8-3) \leq 0$$

$$(x+4)(2x+5) \leq 0$$

$$-4 \leq x \leq -2\frac{1}{2} \quad (A)$$



5. $\lim_{x \rightarrow \infty} \frac{x+1}{x-1}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x} + \frac{1}{x}}{\frac{2x}{x} - \frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{2x}{1} \quad (C)$$

$$= 2x$$

6. $A(6, -4), B(0, 4), P(-3, 8)$

$$m:n = k:1$$

$$-3 = \frac{k \times 0 + 1 \times 6}{k+1}$$

$$-3k - 3 = 6$$

$$-3k = 9 \rightarrow k = -3 \quad (B)$$

7. $26^2 \times 10^2 \times 26^2 = 45697600 \quad (D)$

8. ${}^9C_5 = 126 \quad (C)$

9. $\lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x+1}$
 $= \lim_{x \rightarrow -1} \frac{(x+1)(x+1)}{x+1}$
 $= -1 + 1$
 $= 0 \quad (C)$

10. $\tan 15^\circ = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}}$$

$$= \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{3 - 2\sqrt{3} + 1}{3 - 1}$$

$$= \frac{4 - 2\sqrt{3}}{2}$$

$$= 2 - \sqrt{3} \quad (A)$$

Question 11:

a) 2, 3, 5, 6, 8 and 9.

i) $6 \times 5 \times 4 \times 3 \times 2 = 720$ (1)

or

$6P_5 = 720$ 5 digit numbers.

ii. Half the numbers will be even since there's an equal number of odd and even digits.

$720 \div 2 = 360$ (1)

or

$5P_4 \times 3 = 360$ since there are 3 numbers that are even 2, 6 and 8.

b) i. ${}^{16}C_4 = 1820$ (1)

ii. ${}^{10}C_4 = 210$ (1)

iii. At least one man: All combinations - all women

$= {}^{16}C_4 - {}^{10}C_4$
 $= 1610$

* carry thru
mark given
(1) for (i) - (ii)
even if (i) or (ii)
incorrect

or

${}^6C_1 \times {}^{10}C_3 + {}^6C_2 \times {}^{10}C_2 + {}^6C_3 \times {}^{10}C_1 + {}^6C_4 \times {}^{10}C_0 = 1610$

c) i. Four men and 4 women at a circular table:

No restrictions: $7! = 5040$ (1)

ii. 1st position is fixed men: $3! 4! = 144$. (1)

iii. 4 men and one group of women = 5 groups

$\therefore 4!$

4 women = $3!$

2 women can sit together in $2!$ ways

$\therefore$ total number of arrangements = $4! 3! 2!$
 $= 288$

-2-

2 Br correct
1 Br dealing
with women
seating.

carry thru
mark given if
(i) answered $4! \times 4!$
and (ii) $4! \times 4! \times 2!$

d) i. $\angle AXC = \angle AYC$

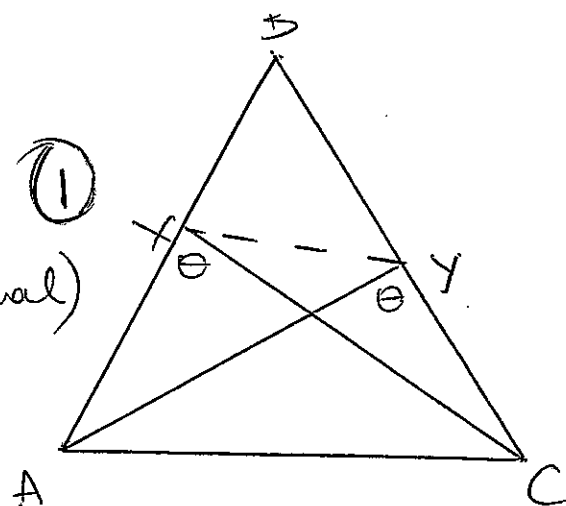
$= \theta$ (given)

$\therefore AXYC$ is cyclic (converse of $\angle$ s in the same segment, AC, are equal) ①

ii. Since $AXYC$ is cyclic,

$\angle BXY = \angle YCA$ (external

$\angle$ in a cyclic quadrilateral = opposite interior $\angle$). ①



e) i. $\cos x - \sin x = A \cos(x + \alpha)$, $0 \leq x \leq 90^\circ$

$A = \sqrt{a^2 + b^2}$

$A = \sqrt{1^2 + 1^2}$

$A = \sqrt{2}$

$\tan \alpha = \frac{b}{a}$

$\tan \alpha = 1$

$\therefore \alpha = 45^\circ$

$\therefore \cos x - \sin x = \sqrt{2} \cos(x + 45^\circ)$

2 mark $\sqrt{2} \cos(x + 45^\circ)$
1 mark either $\sqrt{2}$ or $+45^\circ$.

ii. $\cos x - \sin x = -1$, $0 \leq x \leq 360^\circ$

$\sqrt{2} \cos(x + 45^\circ) = -1$

$\cos(x + 45^\circ) = -\frac{1}{\sqrt{2}}$

2nd and 3rd quadrants.

$\therefore x + 45^\circ = 180^\circ - 45^\circ, 180^\circ + 45^\circ, 45^\circ \leq x + 45^\circ \leq 360^\circ + 45^\circ$

$x + 45^\circ = 135^\circ, 225^\circ$ ①

$x = 90^\circ, 180^\circ$ ① for $0 \leq x \leq 360^\circ$

2 mark for correct answer.

3- 1 mark some correct progress

Question 12:

a) $9^x - 10 \cdot 3^x + 9 = 0$

let $u = 3^x$

then $9^x - 10 \cdot 3^x + 9 = 0$ is equivalent to:

$$(3^x)^2 - 10 \cdot 3^x + 9 = 0$$

$$(3^x)^2 - 10 \cdot 3^x + 9 = 0$$

i.e. $u^2 - 10u + 9 = 0$ - ①

$$(u-9)(u-1) = 0$$

$$u = 9 \text{ or } u = 1$$

$$3^x = 9 \text{ or } 3^x = 1$$

$$x = 2 \text{ or } x = 0$$

①

①

1 for correct u equation
1 for each correct x-value

b) $\therefore \tan 45^\circ = \frac{h}{OC}$ ①

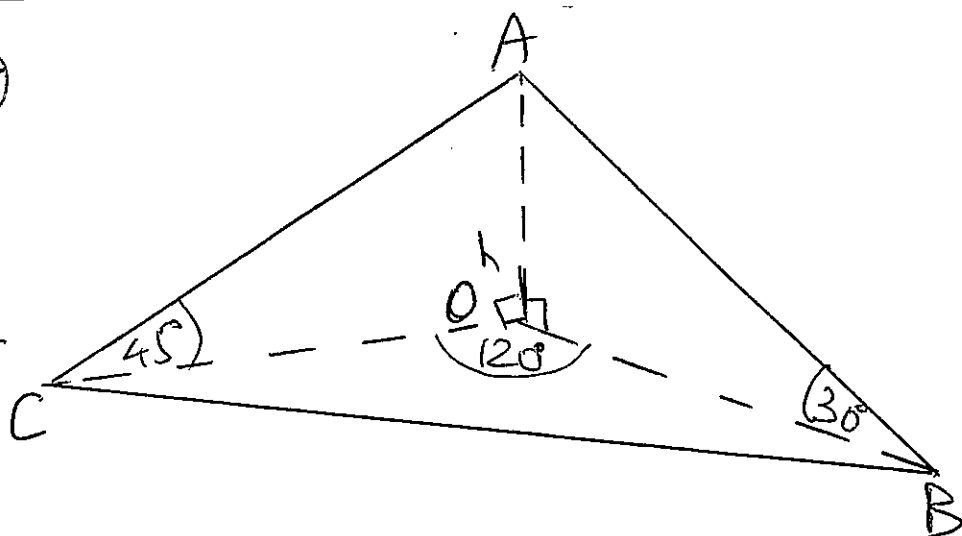
$$1 = \frac{h}{OC}$$

$$\therefore \boxed{OC = h} \quad \text{②}$$

$$\tan 30^\circ = \frac{h}{OB}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{OB}$$

$$\boxed{OB = \sqrt{3}h} \quad \text{③}$$



ii. $BC^2 = OC^2 + OB^2 - 2 \times OC \times OB \times \cos 120^\circ$

$$BC^2 = h^2 + (\sqrt{3}h)^2 - 2 \times h \times \sqrt{3}h \times -\frac{1}{2}$$

1 mark

$$BC^2 = h^2(1 + 3 + \sqrt{3})$$

$$BC^2 = (4 + \sqrt{3})h^2$$

} 1 mark
b.

c) $y = |2x - 3|$

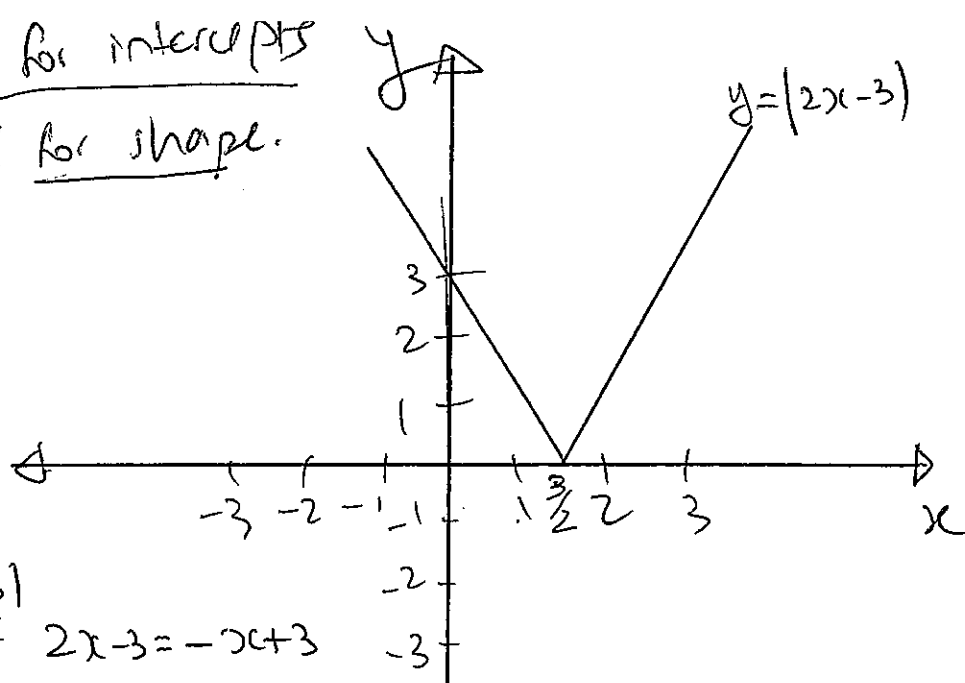
$y = 2x - 3$

or

$y = -2x + 3$

1 for intercepts

1 for shape.



ii. $|2x - 3| \leq |x - 3|$

Solve: $|2x - 3| = |x - 3|$

$2x - 3 = x - 3$ or $2x - 3 = -x + 3$

$x = 0$

or

$3x = 6$

$x = 2$

Test: $x = -1, x = 1, x = 3$

$x = -1$

$|2(-1) - 3| \leq |-1 - 3|$

$|-5| = |-4|$ False.

$x = 1$

$|2(1) - 3| \leq |1 - 3|$

$|-1| \leq |-2|$ True

$x = 3$

$|2(3) - 3| \leq |3 - 3|$

$|3| \leq 0$ False.

$\therefore$ Solution: $0 \leq x \leq 2$

1 for working
1 for result.

d) $A(-4, -3)$, $B(1, 5)$

$m: PA$
 $-3:2$

$$x = \frac{-3 \times 1 + 2 \times -4}{-3 + 2}$$

$$x = \frac{-11}{-1}$$

$$x = 11$$

$$y = \frac{-3 \times 5 + 2 \times -3}{-3 + 2}$$

$$y = \frac{-15 - 6}{-1}$$

$$y = 21 \quad 1 \text{ for each coordinate}$$

$$\therefore P(11, 21)$$

1 mark for significant progress towards answer

e) $\frac{2x+5}{x+1} < 1 \quad x(x+1)^2, x \neq -1$

$$(2x+5)(x+1) < (x+1)^2$$

$$(x+1)^2 - (2x+5)(x+1) > 0$$

$$(x+1)[(x+1) - (2x+5)] > 0$$

$$(x+1)(x+1-2x-5) > 0$$

$$(x+1)(-x-4) > 0$$

$$\boxed{-4 < x < -1}$$

or

Solve $\frac{2x+5}{x+1} = 1$ then test x -values:

$$2x+5 = x+1$$

$$x = -4$$

Test: $x = -5, x = -2, x = 0$

$x = -5$:

$$\frac{2x-5+5}{-5+1} < 1$$

$$\frac{-5}{-4}$$

$$\frac{-5}{-4} < 1 \text{ False.}$$

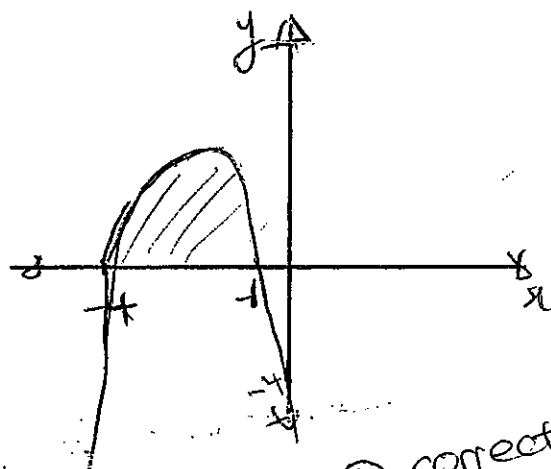
$$\frac{-5}{-4}$$

$x = -2$

$$\frac{2x-2+5}{-2+1} < 1$$

$$\frac{-1}{-2}$$

$$\frac{-1}{-2} < 1 \text{ True.}$$



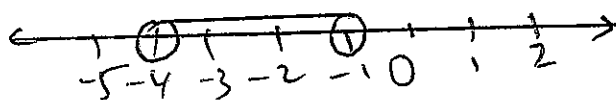
① correct progress having noted $x \neq -1$

$x = 0$

$$\frac{2x+5}{1} < 1$$

$$5$$

$$5 < 1 \text{ False.}$$



$$\therefore -4 < x < -1$$

Question 13:

a) i. $y = x^2 + 2x - 3$ and $y = x^2 + 1$

$$x^2 + 2x - 3 = x^2 + 1$$

$$2x = 4$$

$$x = 2$$

solving simultaneously.

$$y = x^2 + 1 \rightarrow y = 2^2 + 1 \rightarrow y = 5$$

any correct working.

$$\therefore (2, 5)$$

ii. $y = x^2 + 2x - 3$

$$y' = 2x + 2$$

$$\therefore m_1 = 2 \times 2 + 2$$

$$m_1 = 6$$

$$y = x^2 + 1$$

$$y' = 2x$$

$$\therefore m_2 = 2 \times 2$$

$$m_2 = 4$$

1 for gradients

1 for result.

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{6 - 4}{1 + 6 \times 4} \right|$$

$$\tan \theta = \left| \frac{2}{25} \right|$$

$$\therefore \theta = 4.5739 \dots$$

$$\therefore \theta \approx 5^\circ$$

b) $4x^2 - (6+k)x + 1 = 0$

No real zeros $\rightarrow \Delta < 0$

$$b^2 - 4ac < 0$$

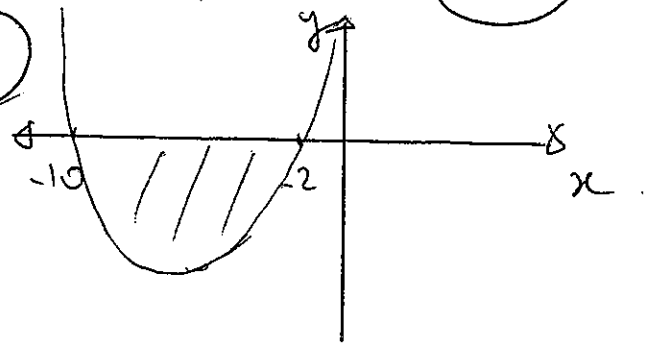
$$[-(6+k)]^2 - 4 \times 4 \times 1 < 0 \quad (1)$$

$$36 + 12k + k^2 - 16 < 0$$

$$k^2 + 12k + 20 < 0$$

$$(k+2)(k+10) < 0$$

$$\therefore -10 < k < -2 \quad (1)$$



c) i. $x \neq \pm 3$ ① $f(x) = \frac{2x^2}{x^2-9}$
 $x=3$ or $x=-3$

ii. $\lim_{x \rightarrow \pm\infty} \frac{2x^2}{x^2 - \frac{9}{x^2}}$

$\lim_{x \rightarrow \pm\infty} \frac{2}{1-0}$

$=2$, y approaches 2 as $x \rightarrow \pm\infty$ ①

iv. Range: $y \leq 0, y > 2$

d) $y = mx$
 $2x - y - 1 = 0$
 $y = 2x - 1$
 $\therefore m_1 = 2$

$\tan 45^\circ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$1 = \left| \frac{2 - m}{1 + 2m} \right|$

$\therefore 1 = \frac{2 - m}{1 + 2m}$ or $-1 = \frac{2 - m}{1 + 2m}$

$1 + 2m = 2 - m$

$-1 - 2m = 2 - m$

$3m = 1$

$-m = 3$

$\therefore m = \frac{1}{3}$ ①

$\therefore m = -3$ ①

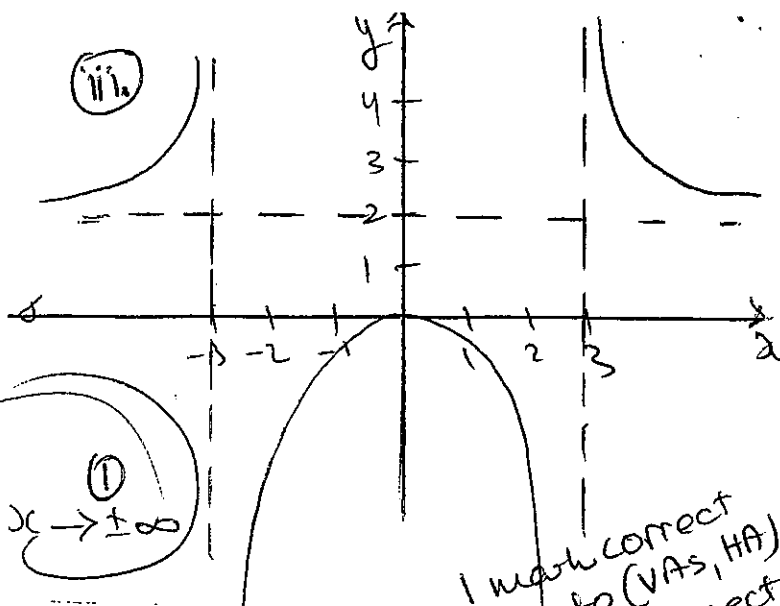
e) $\cos 2x - 3\cos x + 2 = 0$
 $2\cos^2 x - 1 - 3\cos x + 2 = 0$ ←
 $2\cos^2 x - 3\cos x + 1 = 0$
 $(2\cos x - 1)(\cos x - 1) = 0$
 $2\cos x - 1 = 0, \cos x - 1 = 0$

$2\cos x = 1$ or $\cos x = 1$

$\cos x = \frac{1}{2}$

$x = 60^\circ, 360^\circ - 60^\circ$ or $x = 0^\circ, 360^\circ$

$\therefore x = 0^\circ, 60^\circ, 300^\circ, 360^\circ$



1 mark correct into (VAs, HA)
 1 mark correct graph.

1 mark may be awarded for correct progress, eg: correct substitution or one correct solution.

1 for correct substitution for $\cos 2x$

1 for correct $\cos x$ values

1 for solution.